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CIRCULARLY POLARIZED ANTENNA AND INTELLIGENT TERMINAL

Abstract

Provided are a circularly polarized antenna and an intelligent terminal. The circularly polarized antenna includes an annular radiator and a first feed terminal. A first breakpoint is provided on the radiator, and a first capacitor or a first inductor is connected in series with the radiator at the first breakpoint. One end of the first feed terminal is electrically connected to the radiator, and another end of the first feed terminal is electrically connected to a first feed module of a mainboard. When a capacitor or inductor is connected in series with the radiator at the first breakpoint, the resonance frequencies of a first mode and a second mode excited on the radiator which are mutually perpendicular will both change, so that the difference between the resonance phase of the first mode and the resonance phase of the second mode reaches 90° , thereby the circular polarization is realized.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and is a continuation of International Application No. PCT/CN2022/139995, filed on Dec. 19, 2022, the entire content of which is incorporated herein by reference.

FIELD

[0002] The present application relates to the field of antenna technology, more particularly to a circularly polarized antenna and an intelligent terminal.

BACKGROUND

[0003] With the development of intelligent terminals (e.g., mobile phones, wearable devices, computers, etc.), satellite positioning has become one of its main functions. In order to achieve the purpose of satellite positioning and trajectory recording, satellite positioning antennas are indispensable. In order to enhance the transmission efficiency from satellite to the ground (e.g., to enhance penetration and coverage area, etc.), the transmitting antenna of the satellite towards the ground adopts the form of circular polarization. Similarly, in order to enhance the receiving capability of the positioning antenna, the receiving antenna of the terminal device should also adopt the same circularly polarized antenna as the transmitting antenna.

[0004] However, in the related technologies, due to the limitations of size or industrial design of intelligent terminals, it is difficult to implement circularly polarized antennas. Instead, linearly polarized antennas are commonly adopted, which leads to poor satellite positioning performance of intelligent terminals.

SUMMARY

[0005] One objective of the embodiments of the present application is to provide a circularly polarized antenna and an intelligent terminal to solve the problem of poor satellite positioning performance caused by the use of linearly polarized antennas in terminal devices.

[0006] Technical solutions adopted in the embodiment of the present application are as follows:

[0007] In accordance with a first aspect, an embodiment of the present application provides a circularly polarized antenna, including: [0008] an annular radiator, a first breakpoint is provided on the annular radiator, and a first capacitor or a first inductor is connected in series with the annular radiator at the first breakpoint; and [0009] a first feed terminal, one end of the first feed terminal is electrically connected to the annular radiator, and the other end of the first feed terminal is electrically connected to a first feed module of a mainboard.

[0010] In accordance with a second aspect, an intelligent terminal is provided, including the circularly polarized antenna described in the first aspect.

[0011] Beneficial effects of the first aspect provided by the embodiments of the present application include are as follows: in the circularly polarized antenna provided by the embodiments of the present application, an annular radiator and a first feed terminal are included. A first breakpoint is provided on the radiator, a first capacitor or a first inductor is connected in series with the radiator at the first breakpoint, one end of the first feed terminal is electrically connected to the radiator, and the other end of the first feed terminal is electrically connected to a first feed module of a mainboard.

[0012] When a capacitor is connected in series with the radiator at the first breakpoint, the equivalent distributed inductance of the radiator decreases due to the offset effect of the capacitor, and the resonant frequencies of the first mode and the second mode of the radiator both increase, and the resonant current of the first mode of the radiator and the resonant current of the second mode of the radiator are perpendicular to each other. In case that the first breakpoint is located both

in the current weak zone of the first mode and the current strong zone of the second mode, for the first mode, the series-connected capacitor has a relatively small impact on the original current distribution of the radiator, and the increase in the resonant frequency of the first mode of the radiator is relatively small. For the second mode, the series-connected capacitor has a relatively large impact on the original current distribution of the radiator, and the increase in the resonant frequency of the second mode of the radiator is relatively large. By adjusting the position of the first breakpoint and/or the capacitance of the capacitor, the difference between the resonant phase of the first mode and the resonant phase of the second mode is enabled to reach 90° , and then the radiator forms a circularly polarized antenna.

[0013] When the inductor is connected in series with the radiator at the first breakpoint, the inductance of the equivalent distributed inductance of the radiator will increase, the resonant frequency of the first mode and the second mode of the radiator will both decrease, and the resonant current of the first mode of the radiator and the resonant current of the second mode of the radiator are perpendicular to each other. In case that the first breakpoint is located both in the current weak zone of the first mode and the current strong zone of the second mode, for the first mode, the series-connected inductor has a relatively small impact on the original current distribution of the radiator, and the reduction in the resonant frequency of the first mode of the radiator is relatively small. For the second mode, the series-connected inductor has a relatively large impact on the original current distribution of the radiator, and the reduction in the resonant frequency of the second mode of the radiator is relatively large. By adjusting the position of the first breakpoint and/or the inductance of the inductor, the difference between the resonant phase of the first mode and the resonant phase of the second mode is enabled to reach 90° , and then the radiator forms a circularly polarized antenna.

[0014] It thus can be concluded that when the capacitor or inductor is connected in series with the radiator at the first breakpoint, the resonance frequencies of a first mode and a second mode excited on the radiator which are perpendicular to each other will both change, so that the difference between the resonance phase of the first mode and the resonance phase of the second mode reaches 90° , thereby the circular polarization is realized.

[0015] Therefore, the circularly polarized antenna provided in the embodiments of the present application only needs to use one annular radiator, which reduces the number of radiators, reduces the occupied space of the circularly polarized antenna, contributes to the miniaturization design of the intelligent terminal, and improves the satellite positioning performance of terminal devices.

[0016] Beneficial effect of the second aspect provided in the embodiments of the present application is the same as the beneficial effects of the first aspect mentioned above, references can be made to the beneficial effect of the first aspect mentioned above.

Description

BRIEF DESCRIPTION FOR DRAWINGS

[0017] To illustrate technical schemes in the embodiments of the present application more clearly, the drawings required to be used in description of the embodiments or exemplary technologies will be briefly introduced below. Obviously, the drawings in the following description are merely some embodiments of the present application. For ordinary technicians in this field, other drawings may also be obtained based on these drawings without paying any creative efforts.

[0018] FIG. 1 is a schematic diagram of a resonant frequency of a complete circular ring-shaped radiator in a first mode and a second mode provided by an embodiment of the present application;

[0019] FIG. 2 is a schematic diagram of a three-dimensional structure of a circularly polarized antenna provided by an embodiment of the present application;

[0020] FIG. 3 is a schematic structural diagram of a circularly polarized antenna provided by an

embodiment of the present application;

[0021] FIG. **4** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0022] FIG. **5** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0023] FIG. **6** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0024] FIG. **7** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0025] FIG. **8** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0026] FIG. **9** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0027] FIG. **10** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0028] FIG. **11** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application;

[0029] FIG. **12** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application; and

[0030] FIG. **13** is a schematic structural diagram of a circularly polarized antenna provided by another embodiment of the present application.

[0031] In the figures: **100**, radiator; **101**, current strong zone; **102**, current weak zone; **200**, first feed terminal; **300**, first breakpoint; **400**, mainboard; **500**, second breakpoint; and **600**, second feed terminal.

DETAILED DESCRIPTION OF EMBODIMENTS

[0032] To make the objectives, technical solutions and advantages of the present application clearer and more explicit, the present application is further described in detail in conjunction with the accompanying drawings and embodiments. It should be understood that the specific embodiments described herein are merely used to explain the present application and are not intended to limit the present application.

[0033] It should be noted that various steps recorded in a method implementation of the present application may be performed in different orders and/or in parallel. In addition, the method implementation may include additional steps and/or omit the steps shown. The scope of the present application is not limited in this regard. The term “including” and its variations used herein are open inclusions, that is, “including but not limited to”. The term “based on” means “at least partially based on”. The term “an embodiment” means “at least one embodiment”. The term “another embodiment” means “at least one other embodiment”. The term “some embodiments” means “at least some embodiments”. The relevant definitions of other terms will be given in the following description. It should be noted that the concepts of “first” and “second” mentioned in the present application are only used to distinguish different devices, modules or units, and are not used to limit the order or interdependence of the functions performed by these devices, modules or units.

[0034] As shown in FIG. **1**, when there is no breakpoint on the radiator **100**, that is, the radiator **100** is a complete circular ring, a resonant current of the radiator **100** in a first mode (such as pattern A in FIG. **1**) and a resonant current of the radiator **100** in a second mode (such as pattern B in FIG. **1**) are perpendicular to each other, have the same amplitude, and have the same phase (that is, the same resonant frequency). In this case, the radiator **100** is equivalent to a linearly polarized antenna.

[0035] The resonant current of the radiator **100** in the first mode is shown in pattern A of FIG. **1**. Though pattern A of FIG. **1** only shows that the resonant current flows from top to bottom, the

resonant current may also flow from bottom to top. The density of arrows in pattern A of FIG. 1 represents the magnitude of the current. The areas with dense arrows are current strong zones **101**, and the areas outside the current strong zones **101** are current weak zones **102**. Each current weak zone **102** contains a current zero point. It thus can be seen that the radiator **100** in the first mode includes two current strong zones **101** and two current weak zones **102**.

[0036] The resonant current of the radiator **100** in the second mode is shown in pattern B of FIG. 1. Though pattern B of FIG. 1 only shows that the resonant current flows from left to right, and the resonant current may also flow from right to left. The density of arrows in pattern B of FIG. 1 represents the magnitude of the current. The areas with dense arrows are the current strong zones **101**, and the areas outside the current strong zones **101** are the current weak zones **102**. Each current weak zone **102** contains a current zero point. It thus can be seen that the radiator **100** in the second mode includes two current strong zones **101** and two current weak zones **102**.

[0037] As shown in FIG. 2, the circularly polarized antenna includes an annular radiator **100** and a first feed terminal **200**. Herein, a first breakpoint **300** is provided on the radiator **100**, a first capacitor C1 or a first inductor L1 is connected in series with the radiator at the first breakpoint **300**, one end of the first feed terminal **200** is electrically connected to the radiator **100**, and the other end of the first feed terminal **200** is electrically connected to a first feed module of a mainboard **400**.

[0038] Particularly, the radiator **100** is arranged parallel to the mainboard **400**, and there is a certain interval between the radiator **100** and the mainboard **400**. The distance between the radiator **100** and the mainboard **400** may be set according to actual needs, for example, the distance between the radiator **100** and the mainboard **400** may be set to 2 to 5 mm. The mainboard **400** is the main printed circuit board (PCB, Printed Circuit Board) of an intelligent terminal, and the mainboard **400** is integrated with a processor and a corresponding feed module. The radiator **100** is electrically connected to the mainboard **400** through the first feed terminal **200**, thereby forming an antenna structure. The connection point between the first feed terminal **200** and the radiator **100** is called a feed point, and the feed point is arranged at a position where the resonant current of the first mode of the radiator **100** is close to the resonant current of the second mode of the radiator **100**.

Optionally, the feed point is arranged at a position where the resonant current or electric field of the radiator **100** in the first mode and the second mode are equal. The length of the first breakpoint **300** may be set according to actual needs. For example, the length of the first breakpoint **300** is set to 3 to 5 mm, the first capacitor C1 or the first inductor L1 may be arranged on the mainboard **400**, and the first capacitor C1 and the first sensor may also be arranged at the first breakpoint **300**.

[0039] In some embodiments, the radiator **100** may be made of conductive materials including metals, alloys, etc.

[0040] As shown in FIG. 3, when the first capacitor C1 is connected in series with the radiator at the first breakpoint **300** on the radiator **100**, the equivalent distributed inductance of the radiator **100** decreases due to an offset effect of the capacitor, and the resonant frequency of the first mode of the radiator **100** and the resonant frequency of the second mode of the radiator **100** both increase. Herein, the resonant current of the first mode of the radiator **100** and the resonant current of the second mode of the radiator **100** are perpendicular to each other. In case that the first breakpoint **300** is located both in the current weak zone **102** of the first mode and the current strong zone **101** of the second mode, for the first mode, the series-connected first capacitor C1 has a relatively small impact on the original current distribution of the radiator **100**, and the increase in the resonant frequency of the first mode of the radiator **100** is relatively small. For the second mode, the series-connected first capacitor C1 has a relatively large impact on the original current distribution of the radiator **100**, and the increase in the resonance frequency of the second mode of the radiator **100** is relatively large. By adjusting the position of the first breakpoint **300** and/or the capacitance of the first capacitor C1, the difference between the resonance phase of the first mode and the resonance phase of the second mode is enabled to reach 90°, and then the radiator **100**

forms a circularly polarized antenna.

[0041] As shown in FIG. 4, when the first capacitor C1 is connected in series with the radiator at the first breakpoint 300, the line connecting the first feed terminal 200 and a center of the radiator 100 is a first line, the line connecting the first breakpoint 300 and the center of the radiator 100 is a second line, and a counterclockwise direction facing an upper surface of the radiator 100 is a first direction. Along the first direction, the first line and the second line form a first angle α , where $\alpha \in (0, \pi/2) \cup (\pi, 3\pi/2)$; or alternatively, $\alpha = (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.

[0042] Particularly, the line connecting the first feed terminal 200 and the center of the radiator 100 is the first line, that is, the line connecting the feed point (the contact point between the first feed terminal 200 and the radiator 100) and the center of the radiator 100 is the first line. Given that the first breakpoint 300 is quite small in length, the line connecting the first breakpoint 300 and the center of the radiator 100 can serve as the second line. Optionally, the line connecting the center of the first breakpoint 300 and the center of the radiator 100 may serve as the second line.

[0043] When $\alpha = (0, \pi/2) \cup (\pi, 3\pi/2)$, the first breakpoint 300 is located at an intersection region where the current weak zone 102 of the first mode and the current strong zone 101 of the second mode of the radiator 100 converge, i.e., the first capacitor C1 is connected in series with the radiator at the junction area between the current weak zone 102 of the first mode and the current strong zone 101 of the second mode of the radiator 100. The resonant frequency of the first mode of the radiator 100 increases slightly, and the resonant frequency of the second mode of the radiator 100 increases significantly. Eventually, the difference between the resonant phase of the first mode and the resonant phase of the second mode reach 90°, forming a circularly polarized antenna.

[0044] When $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$, the first breakpoint 300 is located at an intersection region where the current strong zone 101 of the first mode and the current weak zone 102 of the second mode of the radiator 100 converge, i.e., the first capacitor C1 is connected in series with the radiator at the junction area between the current strong zone 101 of the first mode and the current weak zone 102 of the second mode of the radiator 100. The resonant frequency of the first mode of the radiator 100 increases significantly, and the resonant frequency of the second mode of the radiator 100 increases slightly. Eventually, the difference between the resonant phase of the first mode and the resonant phase of the second mode reaches 90°, forming a circularly polarized antenna.

[0045] The circularly polarized antenna is a right-hand circularly polarized antenna when $\alpha \in (0, \pi/2) \cup (\pi, 3\pi/2)$. The circularly polarized antenna is a left-hand circularly polarized antenna when $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$. Optionally, $\alpha \in (0, \pi/3) \cup (\pi, 4\pi/3)$; or alternatively, $\alpha \in (2\pi/3, \pi) \cup (5\pi/3, 2\pi)$

[0046] As shown in FIG. 5, when the first inductor L1 is connected in series with the radiator at the first breakpoint 300 on the radiator 100, the equivalent distributed inductance of the radiator 100 will increase, the resonant frequency of the first mode and the resonant frequency of the second mode of the radiator 100 will both decrease, and the resonant current of the first mode of the radiator 100 and the resonant current of the second mode of the radiator 100 are perpendicular to each other. In case that the first breakpoint 300 is located both in the current weak zone 102 of the first mode and the current strong zone 101 of the second mode, for the first mode, the series-connected first inductor L1 has a relatively small impact on the original current distribution of the radiator 100, and the reduction in the resonant frequency of the first mode of the radiator 100 is relatively small. For the second mode, the series-connected first inductor L1 has a relatively large impact on the original current distribution of the radiator 100, and the reduction in the resonant frequency of the second mode of the radiator 100 is relatively large. By adjusting the position of the first breakpoint 300 and/or the inductance of the first inductor L1, the difference between the resonant phase of the first mode and the resonant phase of the second mode is enabled to reach 90°, and then the radiator 100 forms a circularly polarized antenna.

[0047] As shown in FIG. 6, when the first inductor L1 is connected in series with the radiator at the

first breakpoint **300**, the line connecting the first feed terminal **200** and the center of the radiator **100** is the first line, the line connecting the first breakpoint **300** and the center of the radiator **100** is the second line, the counterclockwise direction of the upper surface facing the radiator **100** is the first direction, and along the first direction, the first line and the second line form a first angle α , where $\alpha=(0,\pi/2)\cup(\pi,3\pi/2)$; or alternatively, $\alpha=(\pi/2,\pi)\cup(3\pi/2,2\pi)$.

[0048] Particularly, the line connecting the first feed terminal **200** and the center of the radiator **100** is the first line, that is, the line connecting the feed point (the contact point between the first feed terminal **200** and the radiator **100**) and the center of the radiator **100** is the first line. Given that the first breakpoint **300** is quite small in length, the line connecting the first breakpoint **300** and the center of the radiator **100** can serve as the second line. Optionally, the line connecting the center of the first breakpoint **300** and the center of the radiator **100** may serve as the second line.

[0049] When $\alpha\in(0,\pi/2)\cup(\pi,3\pi/2)$, the first breakpoint **300** is located at the intersection region where the current weak zone **102** of the first mode and the current strong zone **101** of the second mode of the radiator **100** converge, i.e., the first inductor **L1** is connected in series with the radiator at the junction area between the current weak zone **102** of the first mode and the current strong zone **101** of the second mode of the radiator **100**. The resonant frequency of the first mode of the radiator **100** decreases slightly, while the resonant frequency of the second mode of the radiator **100** decreases significantly. As a result, the phase difference between the resonant phases of the first mode and the second mode reaches 90° , forming a circularly polarized antenna.

[0050] When $\alpha\in(\pi/2,\pi)\cup(3\pi/2,2\pi)$, the first breakpoint **300** is located at the intersection region where the current strong zone **101** of the first mode and the current weak zone **102** of the second mode of the radiator **100** converge, i.e., the first inductor **L1** is connected in series with the radiator at the junction area between the current strong zone **101** of the first mode and the current weak zone **102** of the second mode of the radiator **100**. The resonant frequency of the first mode of radiator **100** decreases significantly, while the resonant frequency of the second mode of the radiator **100** decreases slightly. As a result, the phase difference between the resonant phases of the first mode and the second mode reaches 90° , forming a circularly polarized antenna.

[0051] The circularly polarized antenna is a left-hand circularly polarized antenna when $\alpha\in(0,\pi/2)\cup(\pi,3\pi/2)$. The circularly polarized antenna is a right-hand circularly polarized antenna when $\alpha\in(\pi/2,\pi)\cup(3\pi/2,2\pi)$. Optionally, $\alpha\in(0,\pi/3)\cup(\pi,4\pi/3)$ or $\alpha=(2\pi/3,\pi)\cup(5\pi/3,2\pi)$.

[0052] In one embodiment of the present application, a second breakpoint **500** is also provided on the radiator **100**, and a second capacitor **C2** or a second inductor **L2** is connected in series with the radiator at the second breakpoint **500**.

[0053] Particularly, the first breakpoint **300** and the second breakpoint **500** are provided on the radiator **100**, and a capacitor or an inductor may be connected in series with the radiator at both the first breakpoint **300** and the second breakpoint **500**. In this case, the circularly polarized antenna may be implemented in the following three ways.

[0054] In a first implementation, as shown in FIG. 7, the first capacitor **C1** is connected in series with the radiator at the first breakpoint **300**, and the second capacitor **C2** is connected in series with the radiator at the second breakpoint **500**. The first capacitor **C1** and the second capacitor **C2** may be symmetrically arranged with respect to the center of the radiator **100**, or may not be symmetrically arranged with respect to the center of the radiator **100**. The first breakpoint **300** and the second breakpoint **500** are both arranged at the intersection region where the current weak zone **102** of the first mode and the current strong zone **101** of the second mode of the radiator **100** converge, i.e., the first capacitor **C1** and the second capacitor **C2** are respectively connected in series with the radiator at the junction area between the current weak zone **102** of the first mode and the current strong zone **101** of the second mode of the radiator **100**. Optionally, the first breakpoint **300** and the second breakpoint **500** may be respectively arranged at the region where the current in the first mode of the radiator **100** is minimum while in the second mode is maximum. This facilitates the difference between the resonant phase of the first mode and the resonant phase

of the second mode to reach 90° , thereby the radiator **100** forms a circularly polarized antenna.

[0055] The line connecting the first feed terminal **200** and the center of the radiator **100** is the first line, the line connecting the first breakpoint **300** and the center of the radiator **100** is the second line, and a line connecting the second breakpoint **500** and the center of the radiator **100** is a third line. The counterclockwise direction facing the upper surface of the radiator **100** is the first direction. Along the first direction, the first line and the second line form a first angle α , where $\alpha \in (0, \pi/2) \cup (\pi, 3\pi/2)$; or alternatively, $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$. The first line and the third line form a second angle, where $\beta \in (0, \pi/2) \cup (\pi, 3\pi/2)$; or alternatively, $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$. Optionally, $\beta \in (0, \pi/3) \cup (\pi, 4\pi/3)$; or alternatively, $\beta \in (2\pi/3, \pi) \cup (5\pi/3, 2\pi)$.

[0056] The circularly polarized antenna is a right-hand circularly polarized antenna when $\alpha \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$ and $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$.

[0057] The circularly polarized antenna is a left-hand circularly polarized antenna when $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$ and $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.

[0058] In case that $\alpha \in (0, \pi/2) \cup (\pi, 3\pi/2)$ and $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$, the first capacitor **C1** enables the radiator **100** to form a right-hand circularly polarized antenna, and the second capacitor **C2** enables the radiator **100** to form a left-hand circularly polarized antenna. If the capability of the first capacitor **C1** to drive current rotation on the radiator **100** for right-hand circular polarization is stronger than the capability of the second capacitor **C2** to drive current rotation on the radiator **100** for left-hand circular polarization, then the radiator **100** ultimately forms a right-hand circularly polarized antenna. On the contrary, the radiator **100** ultimately forms a left-hand circularly polarized antenna.

[0059] In case that $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$, and $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$, the first capacitor **C1** enables the radiator **100** to form a left-hand circularly polarized antenna, and the second capacitor **C2** enables the radiator **100** to form a right-hand circularly polarized antenna. If the capability of the first capacitor **C1** to drive current rotation on the radiator **100** for left-hand circular polarization is stronger than the capability of the second capacitor **C2** to drive current rotation on the radiator **100** for right-hand circular polarization, then the radiator **100** ultimately forms a left-hand circularly polarized antenna. On the contrary, the radiator **100** ultimately forms a right-hand circularly polarized antenna.

[0060] In a second implementation, as shown in FIG. 8, the first inductor **L1** is connected in series with the radiator at the first breakpoint **300** and the second inductor **L2** is connected in series with the radiator at the second breakpoint **500**. The first inductor **L1** and the second inductor **L2** may be symmetrically arranged with respect to the center of the radiator **100**, or may not be symmetrically arranged with respect to the center of the radiator **100**. The first breakpoint **300** and the second breakpoint **500** are both arranged at the intersection region where the current weak zone **102** of the first mode and the current strong zone **101** of the second mode of the radiator **100** converge, i.e., the first inductor **L1** and the second inductor **L2** are both connected in series with the radiator at the junction area between the current weak zone **102** of the first mode and the current strong zone **101** of the second mode of the radiator **100**. Optionally, the first breakpoint **300** and the second breakpoint **500** may be respectively arranged at the region where the current in the first mode of the radiator **100** is minimum while in the second mode is maximum. This makes it easier to achieve a phase difference of 90° between the resonant phase of the first mode and the resonant phase of the second mode, thereby the radiator **100** forms a circularly polarized antenna.

[0061] The line connecting the first feed terminal **200** and the center of the radiator **100** is the first line, the line connecting the first breakpoint **300** and the center of the radiator **100** is the second line, and the line connecting the second breakpoint **500** and the center of the radiator **100** is the third line. The counterclockwise direction of the upper surface facing the radiator **100** is the first direction. Along the first direction, the first line and the second line form the first angle α , where $\alpha \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$; or alternatively, $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$. The first line and the third line form

the second angle β , where $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$; or alternatively, $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.
[0062] The circularly polarized antenna is a left-hand circularly polarized antenna when $\alpha = (0, \pi/2\pi) \cup (\pi, 3\pi/2)$ and $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$.

[0063] The circularly polarized antenna is a right-hand circularly polarized antenna when $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$ and $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.

[0064] In case that $\alpha \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$ and $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$, the first inductor L1 enables the radiator 100 to form a left-hand circularly polarized antenna, and the second inductor L2 enables the radiator 100 to form a right-hand circularly polarized antenna. If the capability of the first inductor L1 to drive current rotation on the radiator 100 for left-hand circular polarization is stronger than that of the second inductor L2 to drive current rotation on the radiator 100 for right-hand circular polarization, then the radiator 100 ultimately forms a left-hand circularly polarized antenna. On the contrary, the radiator 100 ultimately forms a right-hand circularly polarized antenna.

[0065] In case that $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$ and $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$, the first inductor L1 enables the radiator 100 form a right-hand circularly polarized antenna, and the second inductor L2 enables the radiator 100 form a left-hand circularly polarized antenna. If the capability of the first inductor L1 to drive current rotation on the radiator 100 for right-hand circular polarization is stronger than the capability of the second inductor L2 to drive current rotation on the radiator 100 for left-hand circular polarization, then the radiator 100 ultimately forms a right-hand circularly polarized antenna. On the contrary, the radiator 100 ultimately forms a left-hand circularly polarized antenna.

[0066] In a third implementation, as shown in FIG. 9, the first inductor L1 is connected in series with the radiator at the first breakpoint 300, and the second capacitor C2 is connected in series with the radiator at the second breakpoint 500. The first breakpoint 300 and the second breakpoint 500 are respectively arranged at an intersection region where the current weak zone 102 of the first mode and the current strong zone 101 of the second mode of the radiator 100 converge, i.e., the first inductor L1 and the second capacitor C2 are respectively connected in series with the radiator at the junction area between the current weak zone 102 of the first mode and the current strong zone 101 of the second mode of the radiator 100. Optionally, the first breakpoint 300 and the second breakpoint 500 may be respectively arranged at the region where the current in the first mode of the radiator 100 is minimum while in the second mode is the maximum. This makes it easier to achieve the phase difference of 90° between the resonant phase of the first mode and the resonant phase of the second mode, thereby the radiator 100 forms a circularly polarized antenna.

[0067] The line connecting the first feed terminal 200 and the center of the radiator 100 is the first line, the line connecting the first breakpoint 300 and the center of the radiator 100 is the second line, and the line connecting the second breakpoint 500 and the center of the radiator 100 is the third line. The counterclockwise direction facing the upper surface of the radiator 100 is the first direction. Along the first direction, the first line and the second line form the first angle α , where $\alpha \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$; or alternatively, $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$. The first line and the third line form the second angle β , where $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$; or alternatively, $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.

[0068] The circularly polarized antenna is a left-hand circularly polarized antenna when $\alpha = (0, \pi/2\pi) \cup (\pi, 3\pi/2)$ and $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.

[0069] The circularly polarized antenna is a right-hand circularly polarized antenna when $\alpha = (\pi/2, \pi) \cup (3\pi/2, 2\pi)$ and $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$.

[0070] In case that $\alpha \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$ and $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$, the first inductor L1 enables the radiator 100 to form a left-hand circularly polarized antenna, and the second capacitor C2 enables the radiator 100 to form a right-hand circularly polarized antenna. If the capability of the first inductor L1 to drive the current rotation on the radiator 100 for left-hand circular polarization is stronger than the capability of the second capacitor C2 to drive the current rotation on the radiator 100 for right-hand circular polarization, then the radiator 100 ultimately forms a left-hand

circularly polarized antenna. On the contrary, the radiator **100** ultimately forms a right-hand circularly polarized antenna.

[0071] When $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$ and $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$, the first inductor **L1** enables the radiator **100** to form a right-hand circularly polarized antenna, and the second capacitor **C2** enables the radiator **100** to form a left-hand circularly polarized antenna. If the capability of the first inductor **L1** to drive the current rotation on the radiator **100** for right-hand circular polarization is stronger than the capability of the second capacitor **C2** to drive the current rotation on the radiator **100** for left-hand circular polarization, then the radiator **100** ultimately forms a right-hand circularly polarized antenna. On the contrary, the radiator **100** ultimately forms a left-hand circularly polarized antenna.

[0072] In one embodiment of the present application, at least one third breakpoint is also provided on the radiator **100**, and a third capacitor or a third inductor is connected in series with the radiator at the third breakpoint.

[0073] Particularly, the radiator **100** includes a first breakpoint **300**, a second breakpoint **500** and at least one third breakpoint. Herein, the first capacitor **C1** or the first inductor **L1** may be connected in series with the radiator at the first breakpoint **300**. The second capacitor **C2** or the second inductor **L2** may be connected in series with the radiator at the second breakpoint **500**, and the third capacitor or the third inductor may be connected in series with the radiator at each third breakpoint. By adjusting the positions of the first breakpoint **300**, the second breakpoint **500** and the third breakpoint(s) on the radiator **100**, and/or adjusting the capacitance of the capacitor and the inductance of the inductor connected in series with the radiator at the breakpoint, the difference between the resonant phase of the first mode and the resonant phase of the second mode is enabled to reach 90° , thereby the radiator **100** forms a circularly polarized antenna. For the specific design principle, references may be made to the description of designing the first breakpoint **300** and the second breakpoint **500** on the radiator **100**, which will not be repeated here.

[0074] As shown in FIG. **10**, the circularly polarized antenna also includes a first filter **RC1**. The first filter **RC1** is used to filter signals of other communication frequency bands except a signal of a second communication frequency band. the signal of the second communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second capacitor **C2**. Or alternatively, the signal of the second communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second inductor **L2**.

[0075] Particularly, the first filter **RC1** is used to filter signals of other communication frequency bands except the signal of the second communication frequency band. At this point, the signal of the second communication frequency band is a signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second capacitor **C2**. Or alternatively, the signal of the second communication frequency band is a signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second inductor **L2**.

[0076] In this case, a signal of a first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second capacitor **C2**. Or alternatively, the signal of the first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second inductor **L2**. Or alternatively, the signal of the first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the second capacitor **C2**. Or alternatively, the signal of the first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the

radiator **100**, the first inductor **L1** and the second inductor **L2**.

[0077] Thus, the second communication band signal can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, and the second capacitor **C2**. Or alternatively, the second communication band signal can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, and the second inductor **L2**.

[0078] the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second capacitor **C2**. Or alternatively, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second inductor **L2**. Or alternatively, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the second inductor **L2**. Or alternatively, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the second inductor **L2**.

[0079] It can be concluded that the signal of the first communication frequency band and the signal of the second communication frequency band do not interfere with each other. By adjusting the capacitance of the first capacitor **C1** or the inductance of the first inductor **L1** at the first breakpoint **300**, and/or adjusting the capacitance of the second capacitor **C2** or the inductance of the second inductor **L2** at the second breakpoint **500**, the circular polarization characteristics of the signal of the first communication frequency band and the signal of the second communication frequency band can be adjusted, and finally, the radiator **1000** forms a dual-frequency circularly polarized antenna.

[0080] Exemplarily, the first filter **RC1** is connected in parallel with the first capacitor **C1** (as shown in FIG. **10**), or the first filter **RC1** is connected in parallel with the first inductor **L1** (not shown in the figure). The first filter **RC1** may be a bandpass filter.

[0081] As shown in FIG. **11**, the circularly polarized antenna further includes a second filter **RC2**. The second filter **RC2** is used to filter signals of other communication frequency bands except the signal of the first communication frequency band. the signal of the first communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first capacitor **C1**. Or alternatively, the signal of the first communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first inductor **L1**.

[0082] Particularly, the second filter **RC2** is used to filter signals of other communication frequency bands except the signal of the first communication frequency band. In this case, the signal of the first communication frequency band refers to the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first capacitor **C1**. Or alternatively, the signal of the first communication frequency band refers to the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first inductor **L1**.

[0083] The first filter **RC1** is used to filter signals of other communication frequency bands except the signal of the second communication frequency band. In this case, the signal of the second communication frequency band refers to the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second capacitor **C2**. Or alternatively, the signal of the second communication frequency band refers to the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second inductor **L2**.

[0084] Thus, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first

capacitor C1. Or alternatively, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first inductor L1. the signal of the second communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second capacitor C2. Or alternatively, the signal of the second communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second inductor L2.

[0085] It can be concluded that the signal of the first communication frequency band and the signal of the second communication frequency band do not interfere with each other. By adjusting the capacitance of the first capacitor C1 or the inductance of the first inductor L1 at the first breakpoint **300**, and/or adjusting the capacitance of the second capacitor C2 or the inductance of the second inductor L2 at the second breakpoint **500**, the circular polarization characteristics of the signal of the first communication frequency band and the signal of the second communication frequency band can be adjusted, and finally the radiator **100** forms a dual-frequency circularly polarized antenna.

[0086] Exemplarily, the second filter RC2 is connected in parallel with the second capacitor C2 (as shown in FIG. **11**), or alternatively, the second filter RC2 is connected in parallel with the second inductor L2 (not shown in the figure). The second filter RC2 may be a bandpass filter.

[0087] In one embodiment of the present application, the circularly polarized antenna also includes a third filter. The third filter is used to filter signals of other communication frequency bands except a signal of a third communication frequency band, the first filter is used to filter signals of other communication frequency bands except the signal of the second communication frequency band, and the second filter is used to filter signals of other communication frequency bands except the signal of the first communication frequency band.

[0088] In this case, the signal of the first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor L1 and the third capacitor. Or alternatively, the signal of the first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor L1 and the third inductor. Or alternatively, the signal of the first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor C1 and the third capacitor. Or alternatively, the signal of the first communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor C1 and the third inductor.

[0089] the signal of the second communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second inductor L2 and the third capacitor. Or alternatively, the signal of the second communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second inductor L2 and the third inductor. Or alternatively, the signal of the second communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second capacitor C2 and the third capacitor. Or alternatively, the signal of the second communication frequency band refers to a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second capacitor C2 and the third inductor.

[0090] the signal of the third communication frequency band refers to the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor L1 and the second inductor L2. Or alternatively, the signal of the third communication frequency band refers to the signal of the working frequency band of the

circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the second capacitor **C2**. Or alternatively, the signal of the third communication frequency band refers to the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second inductor **L2**. Or alternatively, the signal of the third communication frequency band refers to the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second capacitor **C2**.

[0091] Thus, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the third capacitor. Or alternatively, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the third inductor. Or alternatively, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the third capacitor. Or alternatively, the signal of the first communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the third inductor.

[0092] the signal of the second communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second inductor **L2** and the third capacitor. Or alternatively, the signal of the second communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second inductor **L2** and the third inductor. Or alternatively, the signal of the second communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second capacitor **C2** and the third capacitor. Or alternatively, the signal of the second communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the second capacitor **C2** and the third inductor.

[0093] the signal of the third communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the second inductor **L2**. Or alternatively, the signal of the third communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first inductor **L1** and the second capacitor **C2**. Or alternatively, the signal of the third communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second inductor **L2**. Or alternatively, the signal of the third communication frequency band can only propagate through the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100**, the first capacitor **C1** and the second capacitor **C2**.

[0094] It can be concluded that the signal of the first communication frequency band, the signal of the second communication frequency band and the signal of the third communication frequency band do not interfere with each other. By adjusting the capacitance of the first capacitor **C1** or the inductance of the first inductor **L1** at the first breakpoint **300**, and/or adjusting the capacitance of the second capacitor **C2** or the inductance of the second inductor **L2** at the second breakpoint **500**, and/or adjusting the capacitance of the third capacitor or the inductance of the third inductor at the third breakpoint, the circular polarization characteristics of the signal of the first communication frequency band, the signal of the second communication frequency band and the signal of the third communication frequency band can be adjusted, and finally the radiator **100** forms a triple-frequency circularly polarized antenna.

[0095] Exemplarily, the third filter is connected in parallel with the third capacitor, or alternatively, the third filter is connected in parallel with the third inductor.

[0096] Similarly, three or more breakpoints may be arranged on the radiator **100**, and a capacitor or

an inductor is connected in series with the radiator at each breakpoint, and the radiator **100** is enabled to form a multi-frequency circularly polarized antenna when a filter is connected in series with the radiator at each breakpoint. The principle of the multi-frequency circularly polarized antenna is the same as the above description, which is not repeated here.

[0097] In another embodiment of the present application, the circularly polarized antenna also includes a third filter. In this case, the first filter only allows the signal of the second communication frequency band and the signal of the third communication frequency band to propagate. The second filter only allows the signal of the first communication frequency band and the signal of the third communication frequency band to propagate. The third filter only allows the signal of the first communication frequency band and the signal of the second communication frequency band to propagate. the signal of the first communication frequency band is the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first inductor **L1**. Or alternatively, the signal of the first communication frequency band is the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the first capacitor **C1**. the signal of the second communication frequency band is the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second inductor **L2**. Or alternatively, the signal of the second communication frequency band is the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the second capacitor **C2**. the signal of the third communication frequency band is the signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the third inductor **L3**. Or alternatively, the signal of the third communication frequency band is a signal of the working frequency band of the circularly polarized antenna formed by the first feed terminal **200**, the radiator **100** and the third capacitor **C3**.

[0098] It can be concluded that the signal of the first communication frequency band, the signal of the second communication frequency band and the signal of the third communication frequency band do not interfere with each other. By adjusting the capacitance of the first capacitor **C1** or the inductance of the first inductor **L1** at the first breakpoint **300**, and/or adjusting the capacitance of the second capacitor **C2** or the inductance of the second inductor **L2** at the second breakpoint **500**, and/or adjusting the capacitance of the third capacitor or the inductance of the third inductor at the third breakpoint, the circular polarization characteristics of the signal of the first communication frequency band, the signal of the second communication frequency band and the signal of the third communication frequency band can be adjusted, and finally the radiator **100** forms a triple-frequency circularly polarized antenna.

[0099] Exemplarily, the third filter is connected in parallel with the third capacitor, or alternatively, the third filter is connected in parallel with the third inductor.

[0100] In addition, a plurality of filters arranged in parallel may be connected at each of the three breakpoints of the radiator, and the plurality of filters allow signals of multiple frequency bands to propagate, thereby achieving a three-frequency circularly polarized antenna.

[0101] Similarly, three or more breakpoints may be arranged on the radiator **100**, and a capacitor or an inductor is connected in series with the radiator at each breakpoint, and at least one filter is connected at each breakpoint, so that the radiator **100** can form a multi-frequency circularly polarized antenna. The principle of the multi-frequency circularly polarized antenna is the same as the above description and will not be repeated here.

[0102] As shown in FIG. **12**, the circularly polarized antenna also includes a second feed terminal **600**, one end of the second feed terminal **600** is electrically connected to the radiator **100**, and the other end of the second feed terminal **600** is electrically connected to a second feed module on the mainboard **400**.

[0103] Particularly, when the radiator **100** is provided with the first breakpoint **300** and the second

breakpoint **500**, the first capacitor C1 or the first inductor L1 is connected in series with the radiator at the first breakpoint **300**, the second capacitor C2 or the second inductor L2 is connected in series with the radiator at the second breakpoint **500**, and the first filter RC1 is connected in parallel with the first capacitor C1 or the first inductor L1, the radiator **100** forms a dual-frequency circularly polarized antenna.

[0104] In case that the circularly polarized antenna includes the first feed terminal **200** and the second feed terminal **600**, the first feed module on the mainboard **400** connected to the first feed terminal **200** may be a GPS module, in which case the first feed module is used to receive a GPS signal of one frequency band, and the second feed module on the mainboard **400** connected to the second feed terminal **600** may be a Bluetooth module or a WiFi module, in which case the second feed module is used to receive Bluetooth or WiFi signals. Thus, the circularly polarized antenna can achieve simultaneous excitation and reception of single-frequency GPS signals and Bluetooth or WiFi signals.

[0105] The positioning method for the connection point between the second feed terminal **600** and the radiator **100** is consistent with the method for determining the location of the first feed terminal **200** described above, which will not be repeated here.

[0106] As shown in FIG. 13, when the radiator **100** is provided with the first breakpoint **300** and the second breakpoint **500**, the first breakpoint **300** is connected in series with the first capacitor C1 or the first inductor L1, the second breakpoint **500** is connected in series with the second capacitor C2 or the second inductor L2, the first filter RC1 is connected in parallel with the first capacitor C1 or the first inductor L1, and the second filter RC2 is connected in parallel with the second capacitor C2 or the second inductor L2, the radiator **100** forms a dual-frequency circularly polarized antenna. Since the frequency of the signal that the antenna can excite or receive is in a multiple frequency relationship, for instance, the antenna may excite or receive signals in the $f_{sub.0}$ frequency band, signals in the $2f_{sub.0}$ frequency band, and signals in the $3f_{sub.0}$ frequency band. Particularly, the frequency of the GPS signal in the L5 band is approximately 1.176 GHz, and the frequency of the Bluetooth signal or the WiFi signal is 2.4 GHz. The frequency of the Bluetooth signal or the WiFi signal is approximately twice the frequency of the GPS signal in the L5 band. Thus, when the circularly polarized antenna can excite or receive the GPS signal in the L5 band, the circularly polarized antenna can also receive the Bluetooth signal or the WiFi signal simultaneously.

[0107] In case that the circularly polarized antenna includes the first feed terminal **200** and the second feed terminal **600**, the first feed module on the mainboard **400** connected to the first feed terminal **200** may be a GPS module, in which case the first feed module is used to receive the GPS signal, and the second feed module on the mainboard **400** connected to the second feed terminal **600** may be a Bluetooth module or a WiFi module, in which case the second feed module is used to receive the Bluetooth or WiFi signal. Thus, the circularly polarized antenna can realize the simultaneous excitation and reception of dual-frequency GPS signals and Bluetooth or WiFi signals.

[0108] The positioning method for the connection point between the second feed terminal **600** and radiator **100** is consistent with the aforementioned method used for determining the location of the first feed terminal **200**, and thus is not repeated here.

[0109] The present application also discloses an intelligent terminal, which includes the circularly polarized antenna described above. Since one annular radiator is used in the circularly polarized antenna, the space occupied by the antenna is reduced. Therefore, the intelligent terminal is more conducive to realizing a miniaturization design.

[0110] The intelligent terminal of the present application may be a mobile phone, a tablet and an intelligent wearable device. The intelligent wearable device may be an intelligent watch, an intelligent bracelet, an intelligent headset or an intelligent pair of glasses.

[0111] The above embodiments are merely used to illustrate rather than limit the technical solutions of the present application. Although the present application is described in detail with reference to

the above embodiments, ordinary technicians in this field should understand that the technical solutions recorded in the above embodiments may still be modified, or some of the technical features therein may be substituted by equivalents. These modifications or substitutions do not make the essence of the corresponding technical solutions deviate from the spirit and scope of the technical solutions of the embodiments of the present application, and thus should all be included within the protection scope of the present application.

Claims

1. A circularly polarized antenna, comprising: an annular radiator, a first breakpoint is provided on the annular radiator, a first capacitor or a first inductor is connected in series with the annular radiator at the first breakpoint; and a first feed terminal, one end of the first feed terminal is electrically connected to the annular radiator, and another end of the first feed terminal is electrically connected to a first feed module of a mainboard.
2. The circularly polarized antenna according to claim 1, wherein a line connecting the first feed terminal and a center of the annular radiator is a first line, a line connecting the first breakpoint and the center of the annular radiator is a second line, a counterclockwise direction of the annular radiator is a first direction, and along the first direction, the first line to the second line forms a first angle α ; wherein, $\alpha \in (0, \pi/2) \cup (\pi, 3\pi/2)$; or alternatively, $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.
3. The circularly polarized antenna according to claim 2, wherein, $\alpha \in (0, \pi/3) \cup (\pi, 4\pi/3)$; or alternatively, $\alpha \in (2\pi/3, \pi) \cup (5\pi/3, 2\pi)$.
4. The circularly polarized antenna according to claim 2, wherein a second breakpoint is further provided on the annular radiator, and a second capacitor or a second inductor is connected in series with the annular radiator at the second breakpoint.
5. The circularly polarized antenna according to claim 4, wherein the line connecting the first feed terminal and the center of the annular radiator is the first line, a line connecting the second breakpoint and the center of the annular radiator is a third line, the counterclockwise direction of the annular radiator is the first direction, and along the first direction, the first line to the third line forms a second angle β ; wherein, $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$; or alternatively, $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.
6. The circularly polarized antenna according to claim 5, wherein, $\beta \in (0, \pi/3) \cup (\pi, 4\pi/3)$; or alternatively, $\beta \in (2\pi/3, \pi) \cup (5\pi/3, 2\pi)$.
7. The circularly polarized antenna according to claim 4, wherein the circularly polarized antenna further comprises a first filter, the first filter is configured to filter signals of other communication frequency bands except a signal of a second communication frequency band, the signal of the second communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal, the annular radiator and the second capacitor, or alternatively, the signal of the second communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal, the annular radiator and the second inductor.
8. The circularly polarized antenna according to claim 7, wherein the first filter is connected in parallel with the first capacitor, or alternatively, the first filter is connected in parallel with the first inductor.
9. The circularly polarized antenna according to claim 7, wherein the circularly polarized antenna further comprises a second filter, the second filter is configured to filter signals of other communication frequency bands except a signal of a first communication frequency band, the signal of the first communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal, the annular radiator and the first capacitor, or alternatively, the signal of the first communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal, the

annular radiator and the first inductor.

10. The circularly polarized antenna according to claim 9, wherein the second filter is connected in parallel with the second capacitor, or alternatively, the second filter is connected in parallel with the second inductor.

11. The circularly polarized antenna according to claim 7, wherein the circularly polarized antenna further comprises a second feed terminal, one end of the second feed terminal is electrically connected to the annular radiator, and another end of the second feed terminal is electrically connected to a second feed module on the mainboard.

12. The circularly polarized antenna according to claim 9, wherein the circularly polarized antenna further comprises a second feed terminal, one end of the second feed terminal is electrically connected to the annular radiator, and another end of the second feed terminal is electrically connected to a second feed module on the mainboard.

13. An intelligent terminal, comprising: a circularly polarized antenna, comprising: an annular radiator, a first breakpoint is provided on the annular radiator, a first capacitor or a first inductor is connected in series with the annular radiator at the first breakpoint; and a first feed terminal, one end of the first feed terminal is electrically connected to the annular radiator, and another end of the first feed terminal is electrically connected to a first feed module of a mainboard.

14. The intelligent terminal according to claim 13, wherein a line connecting the first feed terminal and a center of the annular radiator is a first line, a line connecting the first breakpoint and the center of the annular radiator is a second line, a counterclockwise direction of the annular radiator is a first direction, and along the first direction, the first line to the second line forms a first angle α ; wherein, $\alpha \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$; or alternatively, $\alpha \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.

15. The intelligent terminal according to claim 14, wherein $\alpha \in (0, \pi/3) \cup (\pi, 4\pi/3)$; or alternatively, $\alpha \in (2\pi/3, \pi) \cup (5\pi/3, 2\pi)$.

16. The intelligent terminal according to claim 14, wherein a second breakpoint is further provided on the annular radiator, and a second capacitor or a second inductor is connected in series with the annular radiator at the second breakpoint.

17. The intelligent terminal according to claim 16, wherein the line connecting the first feed terminal and the center of the annular radiator is the first line, a line connecting the second breakpoint and the center of the annular radiator is a third line, the counterclockwise direction of the annular radiator is the first direction, and along the first direction, the first line to the third line forms a second angle β ; wherein, $\beta \in (0, \pi/2\pi) \cup (\pi, 3\pi/2)$; or alternatively, $\beta \in (\pi/2, \pi) \cup (3\pi/2, 2\pi)$.

18. The intelligent terminal according to claim 17, wherein $\beta \in (0, \pi/3) \cup (\pi, 4\pi/3)$; or alternatively, $\beta \in (2\pi/3, \pi) \cup (5\pi/3, 2\pi)$.

19. The intelligent terminal according to claim 16, wherein the circularly polarized antenna further comprises a first filter, the first filter is configured to filter signals of other communication frequency bands except a signal of a second communication frequency band, the signal of the second communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal, the annular radiator and the second capacitor, or alternatively, the signal of the second communication frequency band is a signal of a working frequency band of the circularly polarized antenna formed by the first feed terminal, the annular radiator and the second inductor.

20. The intelligent terminal according to claim 19, wherein the first filter is connected in parallel with the first capacitor, or alternatively, the first filter is connected in parallel with the first inductor.
